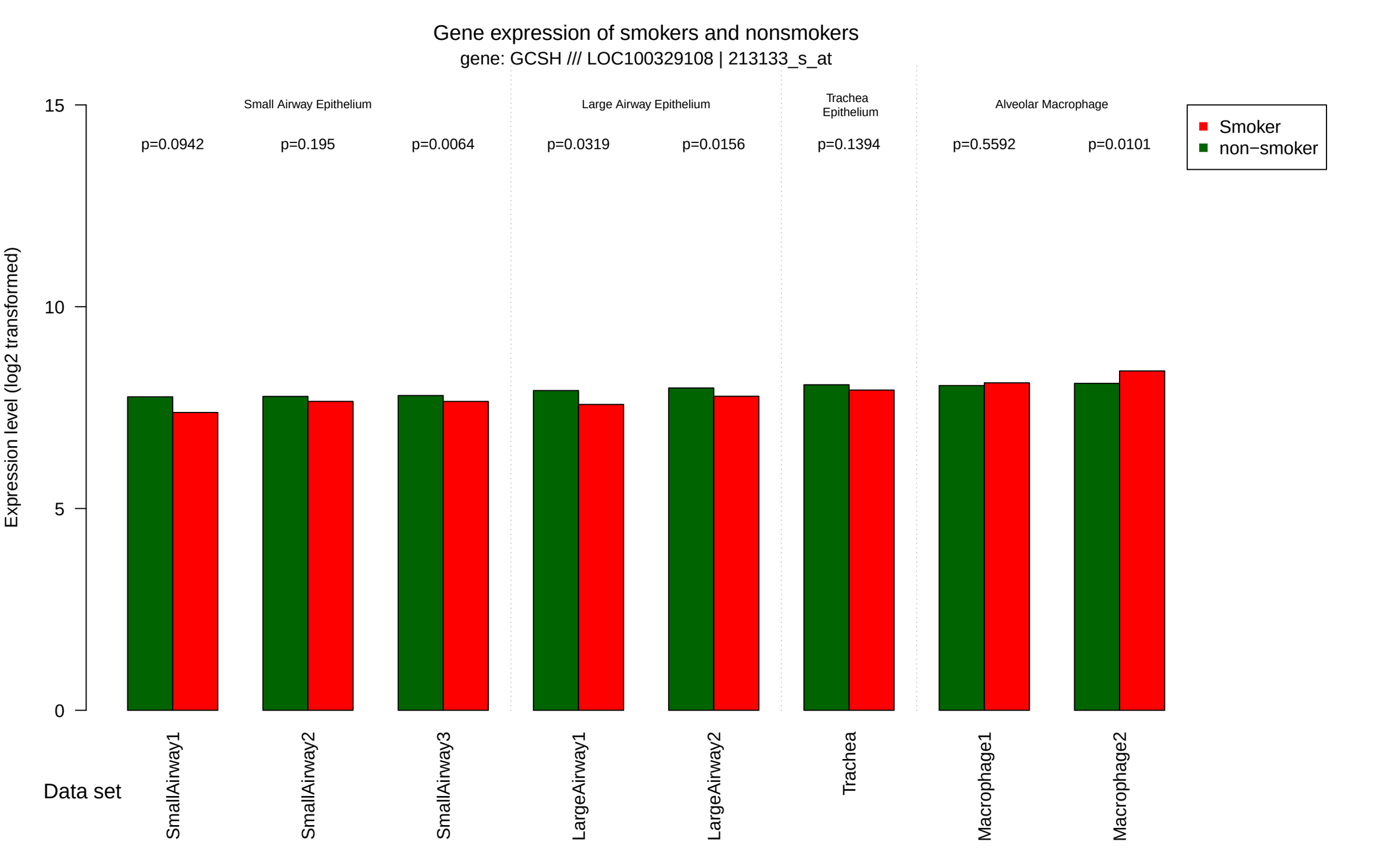
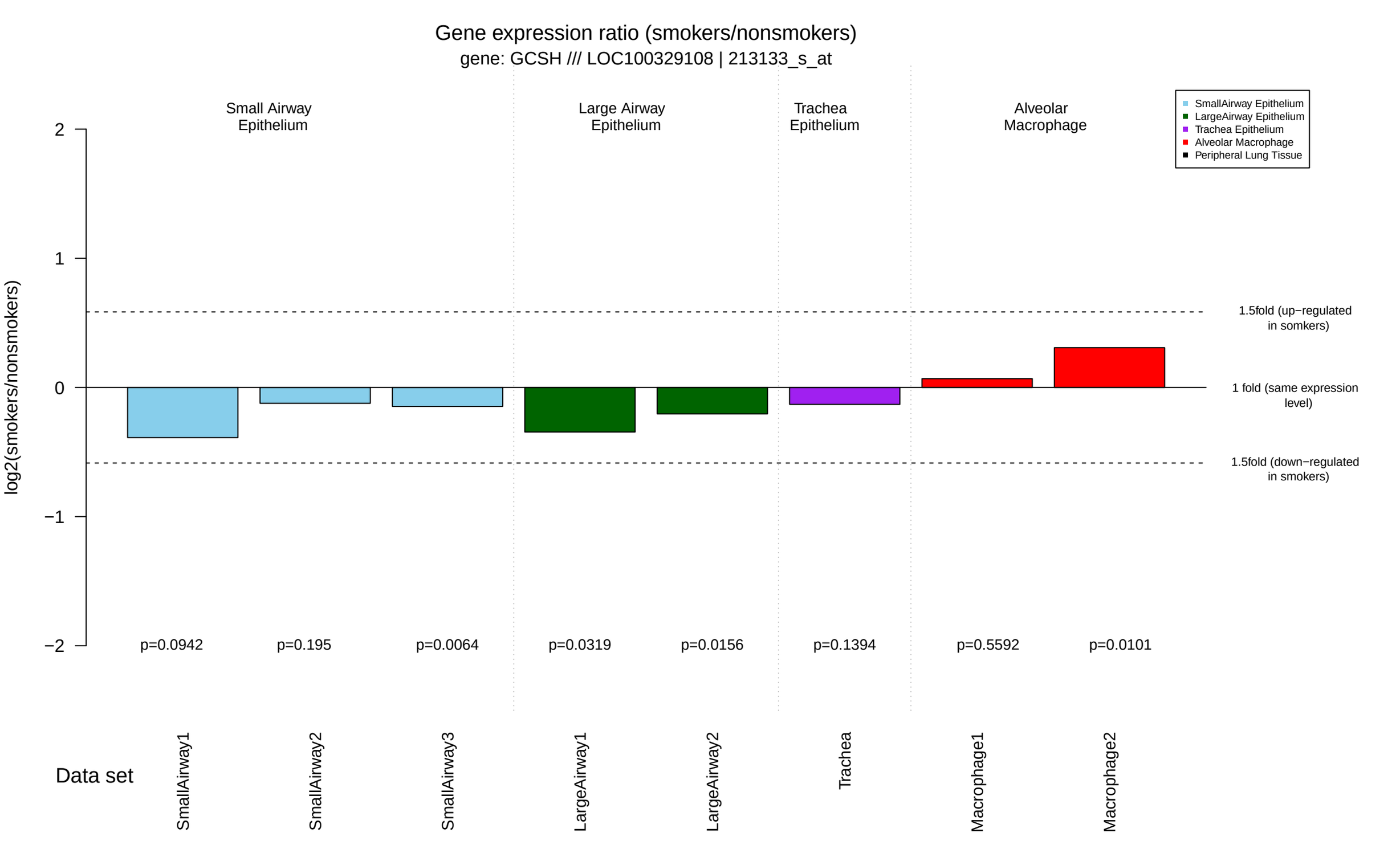
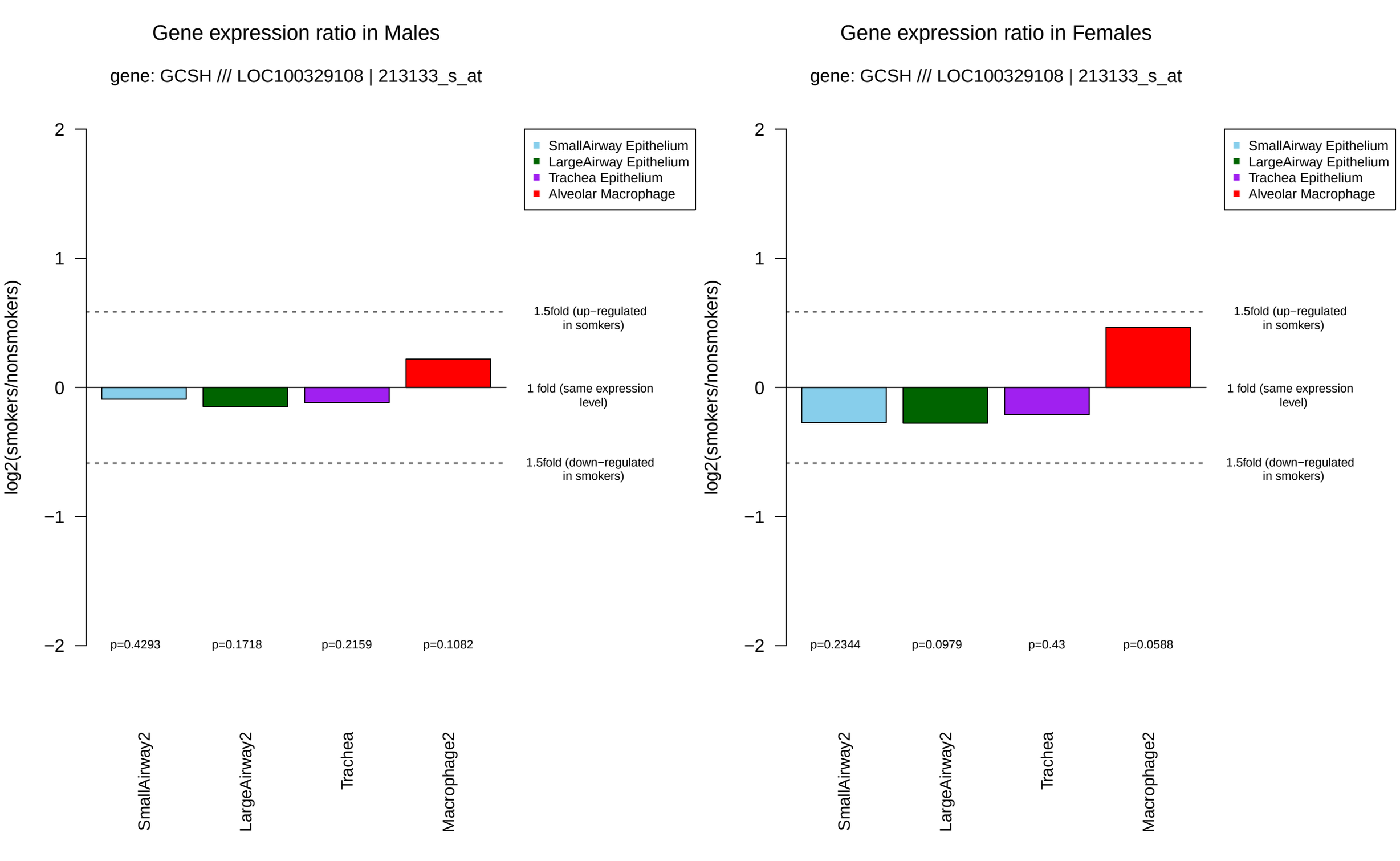
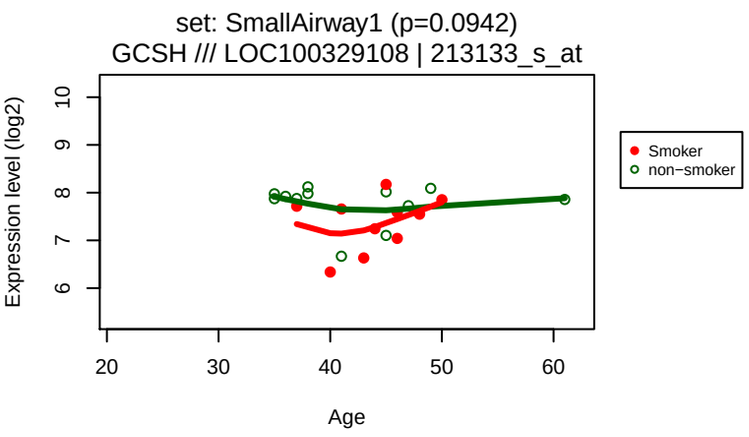
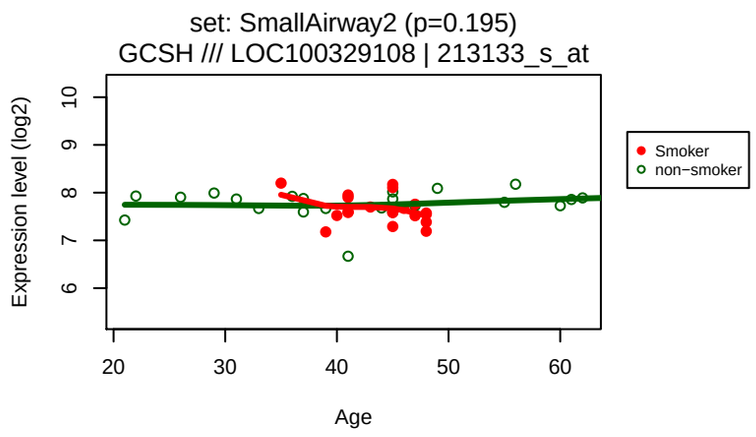
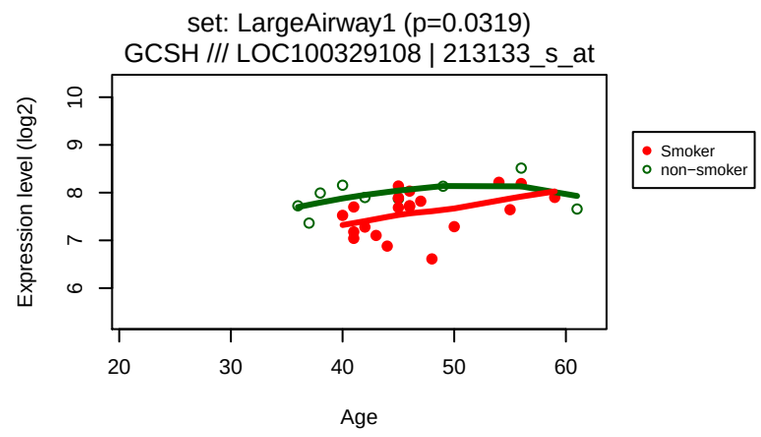

Gene expression of smokers and nonsmokers
gene: GCSH /// LOC100329108 | 213133_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0942
p=0.195
p=0.0064
p=0.0319
p=0.0156
p=0.1394
p=0.5592
p=0.0101
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

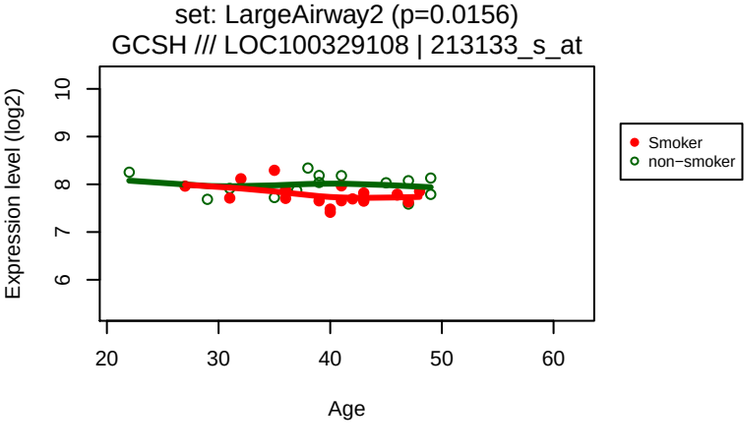

Gene expression ratio (smokers/nonsmokers)
gene: GCSH /// LOC100329108 | 213133_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0942
p=0.195
p=0.0064
p=0.0319
p=0.0156
p=0.1394
p=0.5592
p=0.0101
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

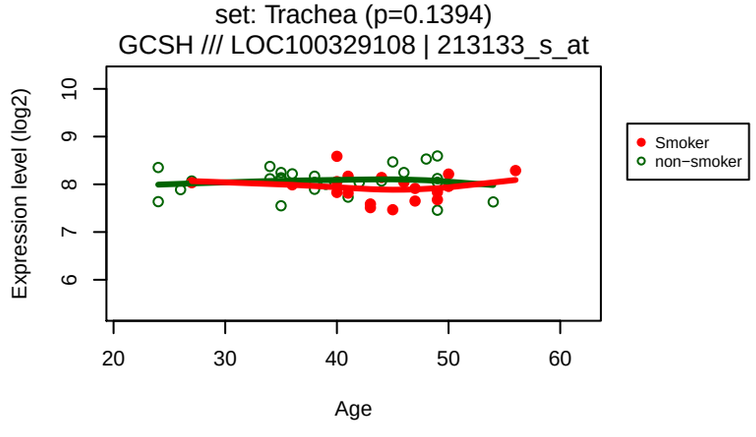

Gene expression ratio in Males
gene: GCSH /// LOC100329108 | 213133_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4293
p=0.1718
p=0.2159
p=0.1082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCSH /// LOC100329108 | 213133_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2344
p=0.0979
p=0.43
p=0.0588
SmallAirway2
LargeAirway2
Trachea
Macrophage2

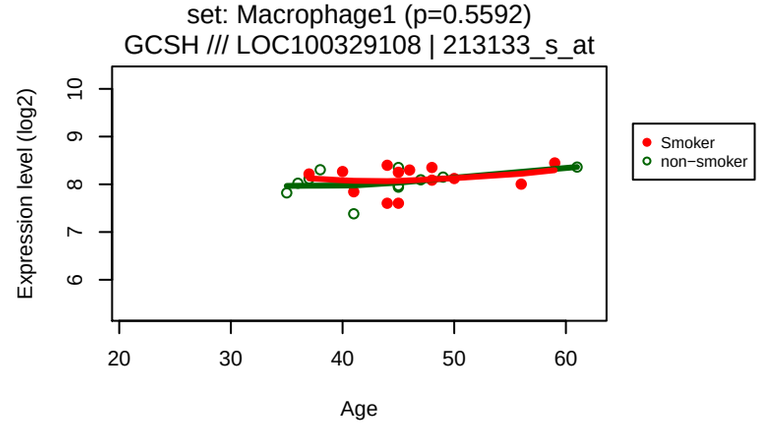

set: SmallAirway1 (p=0.0942)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker

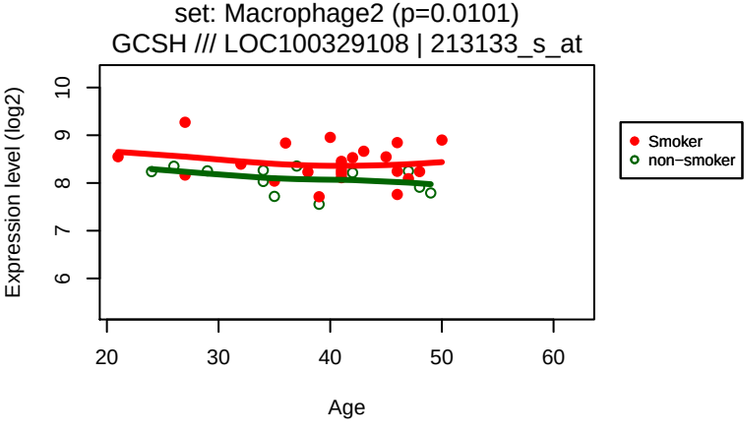

set: SmallAirway2 (p=0.195)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker

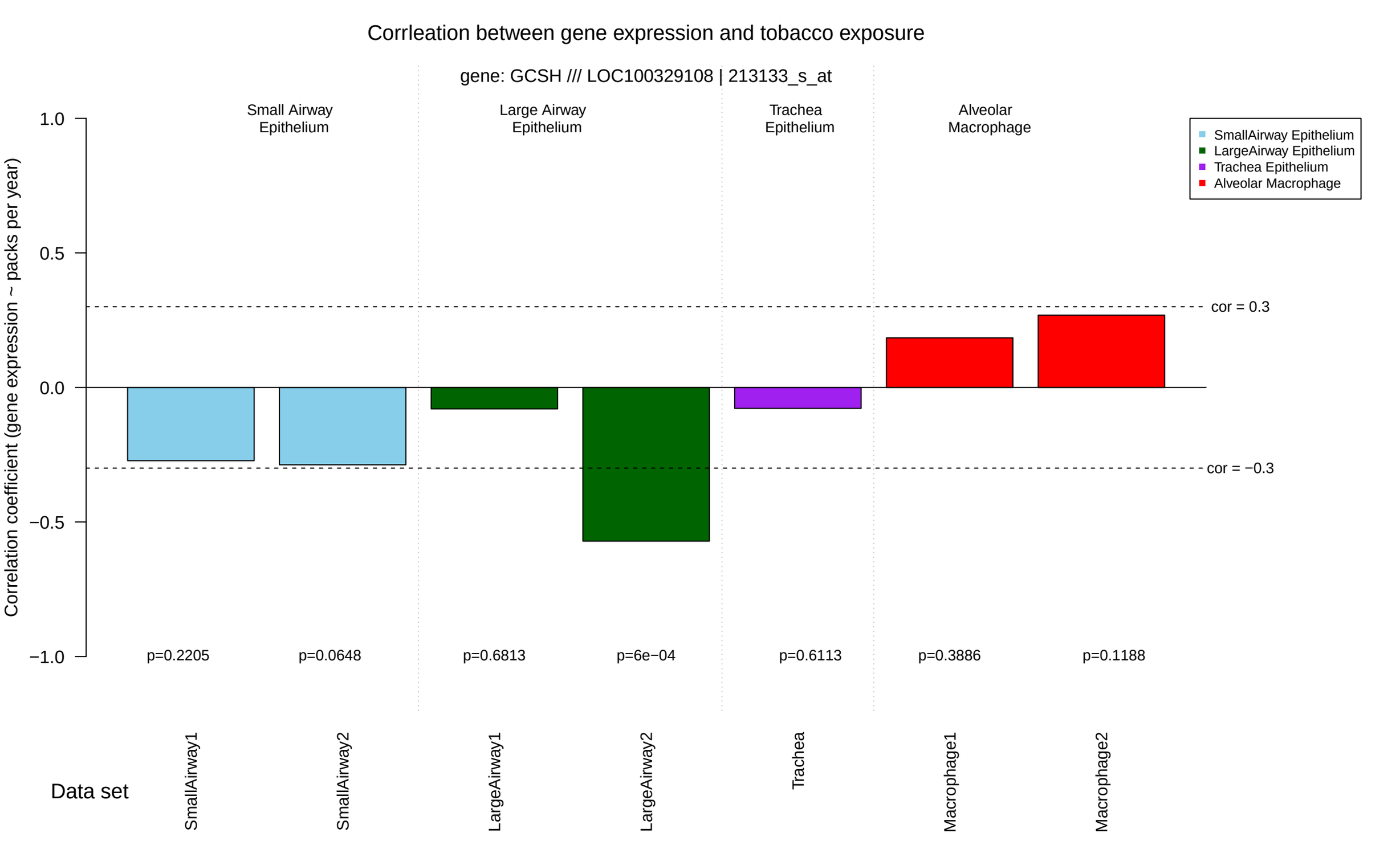

set: LargeAirway1 (p=0.0319)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0156)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1394)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5592)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0101)
GCSH /// LOC100329108 | 213133_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCSH /// LOC100329108 | 213133_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2205
p=0.0648
p=0.6813
p=6e−04
p=0.6113
p=0.3886
p=0.1188
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2